

Rickshesh Manchanda Assignment.

Supender path Exercise

Bug 1 : Incorrect Search results in the backend

File : TaskRepository.java

Issue : The search query didnot group the title and description conditions correctly with the remaining filters. As a result, a task could match the search item without satisfying all the intended filtering conditions.

How i identified it :

I inspected the SQL Query and checked out si how AND and OR conditions interacted.

Fix implemented :

I added parentheses around the title and description search conditions. I also ensured that archived tasks are excluded, status filtering is optional and the results are ordered by creation date in descending order.

How it works :

Parenthesis ensure that the search conditions are evaluated together before the other filters are applied. This makes the search results more consistent with selected filters.

Bug 2 : Incorrect change of page after changing search or status filters.

File : frontend/src/App.jsx.

Issue : when user changed the search query or selected a different status while viewing a later page, the application could remain on that page. This could lead to unexpected or empty results.

How i identified it :

I reviewed the pagination state and manually tested search and status filtering while navigating between pages.

Fix implemented : I reseted the current page to page one whenever the search query or status filter changes.

How it works : The application starts displaying filtered results from the first page instead of retaining an outdated page number.

Bug 3 : Loading indicator not cleared correctly

File : frontend/src/hooks/useTasks.js.

Issue : The loading state needed to be handled reliably when a request succeeded or failed. Otherwise, the interface could continue showing a loading indicator after a failed request.

How I identified it :

I reviewed the task-fetching hook and checked how the loading state was updated during asynchronous requests.

Fix implemented : I used a finally block to set the loading state to false after the request finishes, regardless of its outcome

How it works :

The interface can stop displaying the loading indicator after either a successful or failed request.

Bug 4 : Previous error messages remained visible

File : frontend/src/hooks/useTasks.js.

Issue : An error from a earlier request could remain visible when the application started fetching tasks again.

How i identified : I inspected the error-state mangement in task-fetching logic

Fix implemented : I learned the previous at the beginning of a new fetch operation.

why the fix works :

The interface no longer keeps displaying a stale error while a new request is being processed. Any new Error can be handled seperately.

Bug 5 : Unsafe or invalid pagination and filter parameters.

File : backend/src/main/Java/com/internal/tasktracker/
TaskController.java

Issue : Request parameters such as page number, page size, search text, and status needed validation and normalization before being used by the backend.

How I identified it : I reviewed the controller's request-handling logic and examined how incoming query parameters were processed.

Fix implemented : I added validation for pagination and status parameters, safer pagination handling, trimming of search text, and status normalization. I also removed the artificial query-dependent delay and unnecessary logging.

How it works : Valid inputs are processed more consistently, invalid pagination or status values can be rejected safely, and the unnecessary delays no longer slow down task requests.

Bug 6 : poor handling of loading, error, and empty task states.

File : frontend/src/components/TaskTable.jsx

Issue : The task table needed clearer handling of different states, including loading, request errors, and cases where no tasks matched the filters. Missing task details could also result in incomplete displays.

How I identified it :

I reviewed the component's conditional rendering and manually checked the interface for different task list states.

Fix implemented : I improved the loading error - and empty result displays and added fallback values for missing status, priority and assignee information. I also added a wrapper to support horizontal scrolling

How it works :

Users receive clearer feedback about what the application is doing, and missing task details are handled more gracefully

Bug 7 : Oracle SQL search conditions were not grouped correctly.

File : db/oracle/task-search:package.sql

Issue : The Oracle SQL queries needed proper grouping of title and description search conditions so that filtering behaved consistently.

Fix
~~How I identified~~ : I added parentheses around the title and description conditions in both queries.

~~How I identified~~ : I reviewed the search conditions in both the count query and the query that retrieves the matching tasks.

Why it works : Both queries apply the same logical grouping, helping the total count remain consistent with the retrieved search results. This SQL change has not yet been verified against a running ~~an~~ Oracle database.